TerraTools · IRRI-SAH

Complete User Manual

Tools, input formats, settings, internal parameters and reporting guide

This manual covers **all 16 tools** of TerraTools. For each tool it gives the purpose, the **input files and how they are handled**, the **settings you choose** (with defaults and ranges), the **internal parameters that are fixed in the program**, the outputs, and a **ready-to-edit methods paragraph**. Chapter 19 collects what to write down so that any result can be reported completely and reproduced.

Category	Tools (chapter)
Data Analysis	Phenotypic Data Analysis (2), Mendelian Segregation (3)
Trait Mapping	Convert Alleles (4), Linkage Mapping (5), QTL Mapping (6), QTL-Seq (7), Diversity Analysis (8), GWAS (9), ET-GWAS (10), PyBSA-Seq (11), Genes Extraction (12), Gene Prioritization (13)
Genomic Selection	Clustering & Genomic Selection split (14), Additive Dominance (15), BreedSight Genomic Prediction (16), Optimal Cross Selection (17), Heterosis (18)

Installer version 1.0.0. Statistical analyses are run by R 4.6.1 and Python as bundled with the installer. Parameter values are those of the app as built on 31 October 2026; values marked fixed cannot be changed in the app. Citations should be checked against the originals before submission.

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1. About TerraTools and how to read this manual

TerraTools is a desktop application of IRRI-SAH that gathers the analyses of a plant-breeding programme in one place: phenotype analysis, marker-based mapping (linkage, QTL, GWAS, bulked-segregant analysis), candidate-gene discovery and genomic selection. Each tool has its own tab, takes files you choose, runs the analysis locally (R and Python are bundled) and writes tables, plots and a report into a results folder.

1.1 The tools and how they connect

Step	Tool	Takes	Gives (and what uses it next)
1	Phenotypic Data Analysis	Multi-environment trial data	Adjusted means, heritability, stability – the **phenotype file** for mapping and prediction
2	Mendelian Segregation	Marker genotypes 0/1/2	Distortion test; non-distorted marker list for Linkage Mapping
3	Convert Alleles → Linkage Mapping	Raw allele calls → coded genotypes	Linkage map and the **genotype file for QTL Mapping**
4	QTL Mapping	Genotype file + phenotype CSV	QTL table and **intervals (bp)** for Genes Extraction
5	QTL-Seq / PyBSA-Seq	Bulk sequencing data	QTL regions (bp)
6	GWAS / ET-GWAS	Genotypes + phenotype	MTAs (input for Genes Extraction)
7	Genes Extraction → Gene Prioritization	Intervals / SNPs + GFF3	Gene list → ranked candidate genes
8	Diversity Analysis	Haplotype-block codes	Diversity indices, clusters, tree
9	Clustering → BreedSight / Additive Dominance	SNP matrix	Training/testing split; relationship matrices; genomic predictions
10	Optimal Cross Selection / Heterosis	GEBV + kinship, or parent genotypes	Cross plan; hybrid predictions

1.2 Settings and fixed parameters

Every chapter separates two kinds of numbers:

- **Settings** – what you choose in the app. The tables give the **default**, the allowed **range** and the **effect**.
- **Fixed (internal) parameters** – values built into the program (random seeds, filters, algorithm constants). They are the same for every run, cannot be changed in the app, and still shape the result, so they belong in a methods section.

Rule of thumb for reporting

Report every setting that is not the default, the **values** of thresholds actually used (not only how they were obtained), the sample and marker numbers before and after filtering, the random seed, and the software versions of section 1.4. The QC reports and logs of the tools list what was removed and why.

1.3 General conventions

Topic	Convention
Genotype codes	0 = homozygous for parent 1 (reference), 1 = heterozygous, 2 = homozygous for parent 2 (alternative); −1 or blank = missing (details per tool)
Chromosome names	Chr01, chr1, S1 and 1 are treated as the same chromosome by the tools that match annotations (Genes Extraction, Gene Prioritization)
Marker names	Names such as `chr02_16919033` carry the physical position in bp; several tools use it (marker order, bp conversion of QTL intervals)
File types	CSV, TSV and Excel (.xlsx/.xls) are read; Excel files are read from the first sheet unless the tool lets you choose
Missing data	Never filled silently except where stated (QTL Mapping flanking-marker fill, mean/mode imputation in GWAS, Clustering, Additive Dominance)
Random seeds	Fixed per tool (listed in the chapters), so a run can be repeated exactly
Results folder	Every run writes its tables, plots and a parameter record to the folder you choose; keep it with the manuscript

1.4 Software and package versions (bundled)

Component	Version	Used by
R	4.6.1	Phenotypic, Mendelian, Linkage, QTL, Additive Dominance, Genes Extraction, GWAS (R models)
lme4 / emmeans / agricolae / car	2.0-6 / 2.0.4 / 1.3-7 / 3.1-5	Phenotypic Data Analysis
cluster / factoextra / moments / openxlsx	2.1.8.3 / 2.2.0 / 0.14.1 / 4.2.9	Phenotypic Data Analysis
randomForest / e1071 / xgboost / caret	4.7-1.2 / 1.7-17 / 3.2.1.1 / 7.0-1	QTL Mapping (RF, SVR, XGBoost, hold-out split)
rrBLUP / sommer	4.6.3 / 4.4.6	Additive Dominance
rMVP / GAPIT / mrMLM	1.4.6 / 3.4.0 / 5.0.1	GWAS (FarmCPU / BLINK / mrMLM and FASTmrMLM)
jsonlite / writexl / ggplot2 / glmnet	2.0.0 / 2.0.1 / 4.0.3 / 5.0	Output and plotting
Python packages	numpy, pandas, scipy, scikit-learn, statsmodels, TensorFlow, matplotlib, seaborn, plotly	GWAS, ET-GWAS, QTL-Seq, Diversity, Clustering, BreedSight, OCS, Heterosis

The Python library versions are those of the build; state the TerraTools installer version (1.0.0) and the versions above in the methods.

2. Phenotypic Data Analysis

Analyses multi-environment field trials: adjusted genotype means per environment, variance components and heritability, correlations, principal components, clustering of genotypes and stability across environments. Calculations run in R (file `phenotypic_engine.R`).

2.1 Input

One Excel (.xlsx, .xls) or .csv file, one row per plot. The columns needed depend on the experimental design:

Design	Required columns	Notes
Augmented (default)	`Genotype, Environment, Rep, Block`	Checks are repeated in every block, new entries once. Blocks and reps are treated as fixed
RCBD	`Genotype, Environment, Rep`	Complete blocks; Rep is a random effect
Alpha-lattice	`Genotype, Environment, Rep, Block`	Incomplete blocks nested in Rep; Rep and Block are random effects

- Every other **numeric** column is a trait. Layout columns (`Plot, Row, Column, Range, Year, Season, Entry, Line.type, Remarks`) are not offered as traits unless you tick them.
- At least **2 environments** are needed for heritability, G×E and stability.
- Missing values are allowed; rows with a missing value for a trait are left out of the analysis of that trait.

2.2 Settings in the app

Setting	Default	Range / options	Effect
Experimental design	Augmented	Augmented / RCBD / Alpha-lattice	Chooses the model for the adjusted means (2.3)
Trait(s)	first numeric	any numeric column	Trait(s) analysed; one run covers all ticked traits
Include G×E interaction	off	on / off	Adds the Genotype × Environment term to the model/tables
Analysis mode	All environments	all / specific environments	Which environments enter the means, ANOVA and clustering (stability always uses the full data)
Number of clusters (k)	3	2 – 10	Groups of genotypes in the clustering step
Clustering method	k-means	k-means / hierarchical	See 2.3
Colour scheme	default	default, blue, viridis, plasma, magma	Plots only; does not change numbers

2.3 Internal (fixed) parameters

Parameter	Value	Meaning
R packages	tidyverse, lme4, emmeans, agricolae, car, broom, cluster, factoextra, pheatmap, corrplot, moments, openxlsx	Versions in the bundled runtime: lme4 2.0-6, emmeans 2.0.4, agricolae 1.3-7, car 3.1-5, cluster 2.1.8.3, factoextra 2.2.0, moments 0.14.1, openxlsx 4.2.9
Augmented design	Federer analysis with fixed blocks	`lm(Value ~ Block + Genotype)` with Rep+Block as the block factor (same model as *agricolae::DAU.test*); adjusted means by *emmeans*; entries split into checks and new entries
RCBD / alpha-lattice	mixed model (lme4::lmer)	Genotype fixed; random Rep (RCBD) or Rep and Block within Rep (alpha); means by *emmeans*
Variance components	lmer: (1|Genotype) + (1|Environment) + (1|Genotype:Environment) + random blocks	Needs ≥ 10 rows and ≥ 2 environments
Broad-sense heritability	$H^2 = V_g / (V_g + V_{ge} + V_e)$	Plot-level (single plot) heritability
Entry-mean heritability	$H^2 = V_g / (V_g + V_{ge}/n_e + V_e/(n_e \cdot n_r))$	n_e = environments, n_r = replications; the value to report for line selection. Plot reference lines at 0.3 and 0.6
PCA	*prcomp*, centred and scaled traits	Needs ≥ 10 rows and ≥ 2 traits
Clustering	scaled trait means; k-means *nstart* 25, or hierarchical Ward.D2	Needs n ≥ k + 2 genotypes
Stability statistics	per genotype over environments: mean, CV, ecovalence, range, rank	Ecovalence = Σ (environment mean of the genotype − overall mean)^2; AMMI-type data: Env_Index, GE_Interaction
Correlations	Pearson, pairwise complete	Between traits
Skewness / kurtosis	*moments* package	In the descriptive statistics table

2.4 Outputs

Output	Content
`Phenotypic_Analysis.xlsx`	Descriptive statistics, adjusted means, ANOVA / variance components, heritability, correlations, PCA scores and loadings, cluster membership
`Environment_Analysis.xlsx`	Environment-wise means and stability statistics
Plots	Distribution, boxplots, heritability bars, correlation matrix, PCA biplot, dendrogram / cluster plot, stability plots, heatmaps (PNG)

2.5 What to report

- Design, number of genotypes, checks, environments, replications and blocks; trait definitions and units.
- Model used for means (augmented Federer / RCBD / alpha-lattice) and software (R 4.6.1; lme4, emmeans, agricolae).
- Variance components and **both** heritabilities (plot-level and entry-mean) with the number of environments and replications used.
- Clustering: method, k and the scaling of traits; PCA: variance explained by the first components.

Methods text – Phenotypic Data Analysis

Phenotypic data from [n] genotypes evaluated in [e] environments ([augmented / randomised complete block / alpha-lattice] design with [r] replications) were analysed with the Phenotypic Data Analysis module of TerraTools in R 4.6.1. For the augmented design, adjusted means were estimated by the classical Federer analysis with fixed block effects (equivalent to agricolae::DAU.test; emmeans 2.0.4); for RCBD and alpha-lattice designs, genotype means were estimated by linear mixed models (lme4) with replication [and block within replication] as random effects. Variance components were obtained from a mixed model with random genotype, environment, genotype × environment and block terms, and broad-sense heritability was calculated on a plot basis as $H^2 = V_g/(V_g+V_{ge}+V_e)$ and on an entry-mean basis as $H^2 = V_g/(V_g+V_{ge}/n_e+V_e/(n_e n_r))$. Traits were standardised for principal component analysis (prcomp) and for [k-means (25 random starts) / Ward hierarchical] clustering into [k] groups. Stability across environments was described by the coefficient of variation and the ecovalence of each genotype.

3. Mendelian Segregation

Tests every marker for departure from the Mendelian ratio expected in the population (chi-square), shows distortion by chromosome and writes a filtered marker list. Calculations in R (`mendelian_engine.R`).

3.1 Input

CSV with `Marker`, `Chromosome` and one column per individual coded **0 / 1 / 2** (0 = parent 1, 1 = heterozygous, 2 = parent 2). Missing codes: the strings you list in the settings (default `NA, ., -9`).

3.2 Settings in the app

Setting	Default	Range / options	Effect
Population type	RIL (1:1)	RIL (1:1) / F2 (1:2:1)	Selects the expected ratio and the degrees of freedom
P-value threshold	0.05	0.001 – 0.2	A marker with p below this is called **distorted**
Missing codes	NA, ., -9	comma-separated	Genotype strings read as missing
Keep only markers with a known chromosome	on	on / off	Markers without a chromosome are left out of the chromosome summaries

3.3 Internal (fixed) parameters

Parameter	Value	Meaning
Test	Pearson chi-square, per marker	At least 5 non-missing genotypes are needed
F2 expectation	0.25 : 0.50 : 0.25, df = 2	Counts of 0, 1 and 2 against the expected counts
RIL expectation	0.5 : 0.5, df = 1	Only classes 0 and 2 are tested; heterozygous calls are ignored for the test
Heterozygosity rate	reported	Column *het_rate* (should be near 0.5 in F2 and near 0 in RIL)
Multiple testing	none applied	The threshold is nominal; a stricter value (e.g. 0.001) can be set to account for many markers

3.4 Outputs

Output	Content
`results_full.csv`	All markers: counts per class, chi-square, p-value, distortion call, het_rate
`results_filtered.csv`	Markers **not** distorted (p ≥ threshold) – can be used as input for Linkage Mapping
Plots	$-\log_{10}(p)$ along the markers with the threshold line; share of distorted markers per chromosome

Methods text – Mendelian Segregation

Segregation distortion was tested for each of [M] markers in [N] [RIL / F2] individuals with a chi-square goodness-of-fit test against the expected [1:1 / 1:2:1] ratio (TerraTools, R 4.6.1; markers with fewer than 5 scored genotypes were not tested). Markers with p < [0.05] were considered distorted ([x] markers, [y] %); [the remaining [z] markers were retained for linkage analysis].

4. Convert Alleles

4.1 Input: raw allele calls

One row per marker. Excel (.xlsx/.xls) or text (.csv, .tsv, .txt). Excel workbooks may have several sheets; the sheet that looks most like genotype data is chosen automatically and can be changed.

Marker	Chromosome	P1	P2	Line_001	Line_002	...
m1	1	A:A	G:G	A:G	G:G	...
m2	1	C:C	T:T	C:C	T:T	...

Raw calls are coded against the parents so that Linkage Mapping can read them (parental alleles a_1 = P1, a_2 = P2).

4.2 Settings in the app

Setting	Default	Meaning / how it is chosen
Raw genotype file	–	.csv .tsv .txt .xlsx .xls
Sheet	auto	Workbook sheet with the most recognised columns (Marker, Chromosome, P1, P2); otherwise the one with most columns
Marker column	auto	Looks for Marker, Markers, SNP, SNP_ID, ID, Locus; otherwise the first column
Chromosome column	auto (optional)	Chromosome, Chr, Chrom, LG, Linkage group
P1 column / P2 column	auto	P1, Parent1, Parent 1, Parent_A, PA, Female / P2, Parent2, Parent 2, Parent_B, PB, Male. Must be two different columns
Allele separator	auto	`:` (A:G) / (A/G) | (A|G) none (AG, two letters). Auto = the separator that reads most of the first 50 calls of every column (up to 2000 calls)

4.3 Coding rules (fixed)

Offspring call	Code	Condition
Homozygous for the P1 allele	0	both alleles = P1 allele
Heterozygous	1	one P1 allele and one P2 allele
Homozygous for the P2 allele	2	both alleles = P2 allele
Missing or unexpected	-1	missing call, or an allele combination the parents cannot explain (for example a third allele)

Calls are upper-cased and trimmed. These strings count as **missing**: empty, `-`, `-:-`, `N`, `NA`, `N:N`, `NAN`, `NONE`, `?`, `./.`, `-/-`, `N/N`; a call containing `-` or `N` as an allele is also missing.

4.4 Markers that are excluded (fixed)

Reason	Rule
Invalid parent	the P1 or P2 call is missing or cannot be read
Non-homozygous parent	a parent is heterozygous (the cross cannot be coded 0/1/2 from it)
Non-segregating	both parents carry the same allele, so the marker is not informative

Columns without a single readable genotype (for example *Position* or notes) are not treated as individuals; they are listed as “columns skipped”.

4.5 Outputs

File / sheet	Content
`<name>_coded.xlsx` – Coded_Markers	**First sheet** = the input of Linkage Mapping: Marker, Chromosome, one 0/1/2/-1 column per individual
– Marker_Details	Per marker: parents, number of missing/unexpected calls, status
– Excluded_Markers	Markers left out and the reason
– Summary	Counts: markers read, informative retained, excluded (by reason), individuals, skipped columns
`<name>_coded.csv`	The Coded_Markers table as CSV

To report: markers read, markers coded, markers excluded by reason, number of individuals, percentage of missing/unexpected calls (shown after conversion), allele separator.

5. Linkage Mapping

5.1 Input: coded genotypes

- Column 1 = `Marker`, column 2 = `Chromosome`, columns 3+ = individuals.
- Codes: **0** = homozygous for parent 1, **1** = heterozygous, **2** = homozygous for parent 2, **-1** = missing (letters a / ab / b are also accepted). Only the **first sheet** of a workbook is read.
- Marker names such as `chr02_16919033` carry the physical position (bp); the tool uses it as the reference order and for the direction of the map. A formula in the Chromosome column makes the tool infer chromosomes from marker names.
- Chromosome **0** means “unplaced” and is left out of the map (listed in the removal log).

5.2 Settings in the app

Setting (tab)	Default	Range / options	Effect
Population type (Upload)	F2	F2, RIL	**F2**: three genotype classes (a, ab, b), two gametes per individual. **RIL**: two classes; heterozygous calls are treated as missing, and distances account for selfing (r = R / (2(1-R)))
Missing data threshold (Parameters)	0.30	0 – 1 (step 0.05)	Markers with a fraction of missing genotypes **greater than** this are removed (“Excess missing data”)
Max recombination fraction (r)	0.45	0 – 0.5 (step 0.01)	Two neighbouring markers with estimated r ≥ this are treated as **unlinked**: the interval is counted as **50 cM** and reported as a linkage break
Marker order	Automatic	Automatic / Sort by physical position in the marker names / Use the order of my file	The starting (reference) order. *Automatic* re-orders a chromosome by the position in the marker names only when the file lists it alphabetically (an accidental sort)
Map method	Find the best marker order	Find the best marker order / Keep my marker order	Both estimate distances with all markers of a chromosome together. The first also checks and improves the order from the genotypes
Chi-square filter	Off	Off / On	Removes segregation-distorted markers (“Segregation distortion”). Expected ratio F2 1:2:1, RIL 1:1; markers with fewer than 10 scored genotypes are never removed
Significance level α (if on)	0.05	0.001 – 0.2	Marker removed when its chi-square p < α
Bonferroni correction (if on)	Off	On / Off	α is divided by the number of markers tested (after missing-data and monomorphic filters)

Hidden settings that the app sets for you

Setting	Value used	Effect
Max total map length	100,000 cM	So large that **no scaling** is applied: the estimated distances are kept as they are
Scaling method	proportional	Only used if the map is longer than the limit above (it would be shrunk proportionally)
Plot options (Map Plot, Density & RF)	bin 10 cM; top 60 pairs	Display only; do not change the map. Map plot is saved at 300 dpi

5.3 Internal (fixed) parameters

These values are the same for every run. They define how the map is built and are the details a reviewer will ask for.

Parameter	Value	Meaning
Mapping function	Kosambi	Converts recombination fraction to cM: d = 25 ln[(1+2r)/(1-2r)]

Parameter	Value	Meaning
Genotyping error rate	0.01 (1 %)	Error term of the hidden Markov model used for multipoint estimation
Grouping	chromosome labels of the file (“given”)	Linkage groups are the chromosomes you supplied; groups are not re-formed de novo
Group check	on, markers are removed	A marker whose strongest linkage (independence LOD) is to **another** chromosome is removed (“Linked to another chromosome”)
Group LOD	5	Minimum independence LOD for that check. A group counts as linked only when at least two of its anchor markers (up to 40 per group) reach it, with a 1 LOD margin over the next group
Markers without a chromosome	assigned or left out	Assigned to the group with LOD ≥ 5 (same rules), otherwise removed (“Unassigned”)
Weak-linkage LOD (min LOD)	3	A marker whose strongest LOD with *any* marker of its chromosome is below 3 is removed (“Weakly linked”); applied to chromosomes with at least 4 unique markers
Poor-fit limit	10 cM	A marker is removed if placing it between its neighbours lengthens the map by more than 10 cM (“Poor fit in map”); at most 20 % of a chromosome’s unique markers are removed this way
Identical markers	merged	Markers with exactly the same genotype pattern (within a chromosome) are represented by one locus and share its position
Monomorphic markers	removed	Markers with only one genotype class among the scored individuals
Missing genotypes	not imputed	Each pair of markers uses only the individuals scored at both; individuals with no data at all are dropped
Pairwise statistic	independence LOD, min. 8 individuals	G^2 of the genotype table / (2 ln 10); insensitive to distortion and to linkage phase, so used for grouping
Order search	greedy + 2-opt + segment insertion + random restarts	Minimises the sum of adjacent recombination frequencies (JoinMap logic)
Order polishing	standard effort: windows of 4 markers (all permutations) and 8 markers (single moves, reversed blocks); up to 5 rounds	Keeps the arrangement with the highest multipoint likelihood (Lep-MAP3 “polish” logic). Alternatives built into the engine: fast (3/5), thorough (5/10)
Given-order preference	3 LOD	The physical (marker-name) order is replaced by the data-driven order only if the likelihood is higher by more than 3 LOD. Without readable positions the data-driven order is used
Multipoint estimation	EM, forward-backward, max 40 iterations, tolerance 1e-5	Estimates r between neighbours from all markers of the chromosome together
Suspect-genotype cut-off	P < 0.05	A genotype whose probability given its neighbours is below 0.05 is counted as suspect (Marker Fit table)
Size limits	2500 unique markers (full search); 800 (order comparison); 240 s per chromosome	Larger chromosomes keep the given order; a note is written to the QC report if a limit is reached
First marker	0 cM	Each chromosome starts at 0

5.4 Processing steps (in order)

- **1.** Read the file; marker order handled as chosen (Marker order). Chromosome 0 is removed.
- **2.** Genotypes are standardised (anything other than a / ab / b becomes missing; no imputation). Individuals without any call are dropped.
- **3.** Markers above the missing-data threshold are removed.

- **4.** Monomorphic markers are removed.
- **5.** Optional chi-square filter (with or without Bonferroni).
- **6.** Identical markers are merged; the group check removes markers linked to another chromosome.
- **7.** For every chromosome: weakly linked markers removed; order searched and polished; compared with the physical order; poor-fit markers removed; final distances estimated by multipoint EM and converted to cM (Kosambi); intervals with r ≥ the recombination threshold are set to 50 cM.
- **8.** Results, plots and the QTL input file are written.

5.5 Outputs and QC items to report

Output	Content
QC Report (tab)	Markers in; removed by chromosome 0 / missing data / monomorphic / chi-square / map construction; markers retained; settings text; chromosome lines (markers, positions, cM, log-likelihood, order source, warnings)
Removed markers (tab, `removal_log.xlsx`)	Marker, Chromosome, Reason, Detail. Reasons: Unplaced (chromosome 0), Excess missing data, Monomorphic marker, Segregation distortion (chi-square), Not linked to any group, Linked to another chromosome, Unassigned (no chromosome), Weakly linked, Poor fit in map
Chromosome Summary	Chromosome, Markers, Unique_Positions, Length_cM, Mean_Spacing_cM, Largest_Gap_cM, Linkage_Breaks, Log_Likelihood, Suspect_Genotypes, Order_Source
Marker Fit	Per position: Scored, Suspect_Genotypes, Suspect_Pct – where genotyping errors are likely
`linkage_results.xlsx`	Full report (map, summary, QC, removed markers, recombination)
`retained_markers.xlsx`, `all_markers.xlsx`	Retained markers with positions; all markers with status, position, reason
Plots	`linkage_map.png`, `marker_density.png`, `recombination_heatmap.png`, cytogenetic ideogram
QTL Mapping input (CSV / xlsx)	Genotypes in the linkage order with chromosome and cM rows; original calls, so missing stays missing

Reading the QC report

Total map length is the sum of *Length_cM*. A **linkage break** means a gap with r ≥ the threshold, counted as 50 cM – map length is then an over-estimate for that chromosome, so mention it. *Order_Source* tells whether the data-driven or the physical order was kept. A large share of “suspect” genotypes points to genotyping errors or a wrong marker position.

6. QTL Mapping

6.1 Input: genotype CSV

Row	Content	Example
1	Marker names (first cell = label)	Marker, m1, m2, m3 ...
2	Chromosome of each marker	Chromosome, 1, 1, 2 ...
3	Map position in cM	Position, 0.0, 12.4, 0.0 ...
4 +	One row per individual: ID, then 0 / 1 / 2	Line_001, 0, 1, 2 ...

Any negative code (-1, -9) or blank / non-numeric cell is read as missing. The easiest way to obtain this file is the *QTL Mapping input (CSV)* button on the Linkage Mapping *Results* tab.

6.2 Input: phenotype CSV

First column = individual ID (must match the genotype file), further columns = numeric traits (header = trait name). At least **8** individuals must be present in both files; individuals with a missing value for the chosen trait are left out of that analysis.

6.3 Input: linkage map (optional)

The Linkage Mapping workbook (`linkage_results.xlsx`, sheet “Linkage Map”) or retained-markers list. It is used only to place the QTL intervals in base pairs (section 6.8); the scan itself uses the cM positions of the genotype file.

6.4 Settings in the app

Setting (Analysis tab)	Default	Range	Effect
Genotype CSV / Phenotype CSV / Linkage map	–	–	Files of sections 6.1–6.3. Linkage map is optional (bp intervals)
Trait	first trait	column of the phenotype file	Trait analysed in this run
Mapping method	CIM	SIM, CIM, MQM, RF, SVR, XGBOOST	See 6.6. SIM/CIM/MQM give true LOD; RF/SVR/XGBoost give an importance score
Significance threshold	Fixed LOD threshold	Fixed LOD / Genome-wide permutation	How the LOD cut-off is set
LOD threshold (fixed)	2.5	0.5 – 10 (step 0.1)	Used when the mode is fixed. Typical reporting values are 3 (single scan) or the permutation value
Permutations	1000	100 – 10000	Number of phenotype shuffles (SIM / CIM / MQM only)
Genome-wide α	0.05	0.01 – 0.2	Threshold = (1 − α) quantile of the permuted genome-wide maximum LOD
Peak p-value cut-off	0.05	0.000001 – 1	A peak must also have a nominal p-value (F-test) at or below this
Peak separation (LOD drop)	2	0.5 – 5	Two peaks on one chromosome are separate QTL only if the LOD curve falls at least this far below the lower peak between them
Support interval – LOD drop	1.5	0.3 – 3	Width of the QTL region: markers where LOD is within this drop of the peak (1.5 ≈ 95 %)
CIM cofactors (max)	5	1 – 20	Maximum number of background markers in CIM (CIM only)
CIM window (cM)	10	1 – 50	Cofactors within this distance of the tested position are left out of the model (CIM only)
Hold-out fraction	0	0 – 0.4 (step 0.05)	0 = all individuals. Otherwise that fraction is held out and the scan uses only the rest (stratified by trait value)

Mapping threshold: which mode should I report?

Genome-wide permutation (α = 0.05) is the one to use for publication: it adapts to the trait, population size and genome. A fixed LOD (for example 3) ignores that and can allow "ghost" peaks – in the training data a fixed LOD of 3 gave two extra peaks that the permutation threshold (3.54) removed. State the threshold **value** and how it was obtained.

6.5 Internal (fixed) parameters – data handling

Parameter	Value	Meaning
Random seed	42	Set before the hold-out split, the permutations and the machine-learning scans – runs are reproducible
Missing genotypes	filled from flanking markers (on)	Each missing call gets its most probable class given the nearest observed marker on each side of the same individual (Kosambi distances; 2-state for RIL / backcross, 3-state for F2). Count is shown in the results line
Marker filter	≥ 2 genotype classes	Markers with a single class are dropped ("near-zero variance"); a marker needs ≥ 8 scored individuals to be tested
Genotype model	1 term for 2 classes, 2 terms (additive + dominance) for 3 classes	More than 5 distinct values: treated as a continuous dosage
Samples	intersection of genotype and phenotype IDs	At least 8 required; those without a value for the trait are removed

6.6 Internal (fixed) parameters – mapping methods

Method	Model and fixed settings
SIM	Single-marker regression (Haley & Knott 1992). **LOD = (n/2) $\log_{10}(RSS_0 / RSS_1)$**; P-value from the F-test of the added genotype terms. No cofactors
CIM	As SIM plus background cofactors chosen by **forward selection with backward elimination**: enter at p < 0.01, drop at p > 0.05; candidates screened by correlation with the residuals (best 30 tested); cofactors closer to each other than the window are not both selected; cofactors within the window (same chromosome) of the tested position are dropped from the model (Zeng 1994). Number and window are set in the app
MQM	Same machinery with a larger background model: up to **15** cofactors, enter p < 0.01, drop p > **0.02**, window **25 cM**. The CIM cofactor / window settings do not apply to MQM
RF	Random forest, **500 trees**; permutation importance (%IncMSE); scaled to a 0 – 10 score
SVR	Support-vector regression, radial kernel, scaled inputs; permutation importance (increase in squared error); scaled to a 0 – 10 score
XGBOOST	Gradient boosting: **100** rounds, max depth 3, learning rate (eta) 0.1, subsample 0.8, column sample 0.8; gain importance; scaled to a 0 – 12 score

RF / SVR / XGBoost are not LOD

Their "LOD" is an importance score rescaled to a LOD-like range so the same peak-calling code can be used; the P-value is only 10^{-score}. Do not report them as likelihood LOD, and the permutation threshold is not available for them (the typed LOD threshold is used). Use them to confirm regions found by SIM / CIM / MQM.

6.7 How a QTL is called (fixed rules)

- **Peak:** a local LOD maximum on a chromosome with LOD ≥ threshold *and* nominal P ≤ the peak p-value cut-off.
- **Separate peaks:** a lower peak is a separate QTL only if the curve between it and the higher peak falls at least *peak separation* below it (the larger of peak separation and support LOD drop is used); otherwise it is a shoulder of the higher peak.

- **Support interval** (Lander & Botstein 1989; R/qtl *lodint*): walk left and right from the peak while LOD ≥ peak − LOD drop. The first marker outside on each side is the **flanking marker** (the region is expanded to markers). *CI_Start_cM* / *CI_End_cM* are the interpolated positions where the curve crosses peak − LOD drop.
- **Bayes credible interval** (95 %; SIM / CIM / MQM only): posterior proportional to 10^{LOD} × marker spacing, within the peak's own domain.
- **Genome-wide P** (permutation mode): (number of permuted maxima ≥ observed LOD + 1) / (permutations + 1).
- **Permutation threshold:** phenotype shuffled, the largest single-marker LOD of the genome recorded, repeated; threshold = (1 − α) quantile (type 7). For CIM and MQM the null comes from single-marker regression – a standard approximation – and this is reported in the threshold note.

6.8 Outputs

Output	Content
QTL table (`qtl_significant_<M>.csv`)	One row per QTL: QTL_ID, Chromosome, Left / Peak / Right marker and position, Interval_Size_cM, CI_Start_cM, CI_End_cM, Bayes_Start_cM, Bayes_End_cM, N_Markers_In_Region, LOD, P_Value, Genomewide_P, PVE_Marker_Percent, PVE_CIM_Percent, Additive_Effect, SE_Additive, CI_Lower, CI_Upper, Dominance_Effect, Dominance_Ratio, Dominance_Class, N_Cofactors, N_Samples, Method, Threshold_LOD, LOD_Drop
Full scan (`qtl_full_scan_<M>.csv`)	Every marker with LOD, P, effects and the QTL it lies in – for your own plots
Manhattan plot, ideogram	`manhattan_<M>.png` (one panel per chromosome, threshold line, QTL shaded), `ideogram_<M>.png` (bar = interval, dot = peak)
QTL intervals (bp)	`qtl_intervals_<M>.csv`: QTL_ID, Chromosome, QTL_Start, QTL_End – input of Genes Extraction / Gene Prioritization

Effect columns

Column	Definition
Additive_Effect (a)	Half the difference between the two homozygote classes, in trait units. Positive = the allele of the higher-coded parent (genotype 2 / "b") raises the trait. For CIM / MQM it is estimated with the background cofactors in the model
SE_Additive, CI_Lower, CI_Upper	Standard error of a; 95 % interval = a ± 1.96 SE
Dominance_Effect (d), Dominance_Ratio	d = heterozygote deviation from the homozygote mean; ratio = |d / a|
Dominance_Class	|d/a| > 1 Overdominance; > 0.5 Dominance; > 0.2 Partial dominance; otherwise Additive (F2 only)
PVE_Marker_Percent	Single-marker R^2 (% of trait variance)
PVE_CIM_Percent	Variance the marker explains *over and above* the cofactors (% of total); equals PVE_Marker for SIM

Converting the interval to base pairs

The mapping works in cM, gene annotations in bp. The confidence interval (CI_Start_cM – CI_End_cM) is converted with the markers whose **name** carries a physical position (or those of the linkage map file): the two ends are placed by linear interpolation between neighbouring markers, and every marker inside the interval is kept so that a map running against the genome still gives a valid span. A chromosome with fewer than two such markers cannot be converted; its QTL are listed as "not in the interval file".

6.9 Compare Models and Common QTL

Compare Models runs the selected methods (default SIM, CIM, MQM) with the same settings and merges overlapping QTL regions: Region_ID, flanking markers, *N_Models_Detected*, *Models_List*, Mean / Max LOD, Mean PVE, and physical positions (Mb) read from marker names with an *Order_Check* (whether the map runs with or against the genome). **Common QTL** keeps regions found by at least *Minimum models* (default 2) methods. A region found by several independent methods is the most reliable result to report.

7. QTL-Seq

Bulked-segregant analysis from sequencing data: SNP-index of a high and a low bulk, Δ(SNP-index), and confidence intervals from simulation (Takagi et al. 2013). Inputs are either BAM files (read counting) or a VCF with the bulks as samples.

7.1 Input

- **VCF mode**: a VCF/VCF.gz with the high bulk, low bulk and (for the index parent) the parent(s) as samples; give the sample names.
- **BAM mode**: BAM files of the bulks and parents plus a reference; reads are counted at the polymorphic sites. Counting uses samtools if found, otherwise pysam, otherwise a built-in reader (engine *auto*).

7.2 Settings in the app

Setting	Default	Range / options	Effect
Input mode	BAM	BAM / VCF	Source of allele counts
Index parent	parent 2	parent 1 / parent 2	The SNP-index counts the allele of this parent
Population	F2	F2 / RIL / BC	Used by the simulation (7.3)
Individuals in high / low bulk	20 / 20	≥ 1	Bulk sizes for the simulation
BC recurrent parent	non-index	index / non-index	Backcross only: which parent is the recurrent one
Min. depth	7	≥ 1	Minimum read depth per bulk at a SNP; max depth 0 = no upper limit
Min. SNP-index filter	0.3 (on)	0 – 1	At least one bulk must reach this index (removes uninformative SNPs)
Min. parent depth / genotype quality	7 / 30		Quality of the parent calls used to pick informative SNPs
Min. mapping / base quality	20 / 20		BAM mode read filters
Min. alternative reads / fraction	2 / 0.05		Evidence needed to call the non-reference allele
Base error	0.01		Sequencing error used in the simulation
Window size / step	2,000,000 bp / 10,000 bp		Sliding window for the smoothed index
Min. SNPs per window	10		Windows with fewer SNPs are not drawn
Simulation replicates	10,000		Number of simulated bulks per depth
Random seed	12345		Makes the simulated intervals reproducible
Depth for the simulation	auto	auto / fixed (20)	Use the observed depths, or a fixed depth
Confidence-interval method	SNP-average	SNP-average / window	How the interval is shown against the Δ(SNP-index) curve
Minimum QTL level	90 %	90 / 95 / 99	Smallest confidence level at which a region is called a QTL
Merge gap	100,000 bp		Significant windows closer than this are merged into one region
Interval basis	top level	top level / window extent / window midpoints	How the QTL start and end are defined
Chromosomes	all	list	Restricts the analysis

7.3 Internal (fixed) parameters

Parameter	Value	Meaning
SNP-index	alt-allele reads of the index parent / total reads, per bulk	Δ(SNP-index) = SNP-index$_{high}$ − SNP-index$_{low}$
Null model	binomial sampling of the bulk and of the reads	F2: Binomial(2n, ½); RIL: Binomial(n, ½); backcross variants; read counts Binomial(depth, allele frequency); depth capped at 200 for the simulation
Confidence intervals	two-sided 90 / 95 / 99 % quantiles of the simulated Δ	Per depth, then smoothed with the same window as the data
Multiple testing	none across windows	The intervals are per window; mention this when reporting
Record of parameters	`analysis_parameters.json`	Written with every run – attach to the report

7.4 Outputs

Output	Content
Tables	`SNP_index_per_position.csv`, `delta_SNP_index.csv`, `all_sliding_windows.csv`, `significant_windows.csv`, `significant_QTLs.csv`, `simulation_confidence_intervals.csv`
Plots	`high_bulk_SNP_index.png`, `low_bulk_SNP_index.png`, `QTLseq_genomewide_plot.png` (and .pdf), per-chromosome plots, `simulation_distribution.png`
Report	`QTLseq_analysis_report.pdf` and `analysis_parameters.json`

Methods text – QTL-Seq

Bulked segregant analysis followed the QTL-seq approach (Takagi et al. 2013). The SNP-index of the high ([n] individuals) and low ([n] individuals) bulk was the proportion of reads carrying the [parent 2] allele; SNPs with a depth below [7] in either bulk, a mapping/base quality below [20] or poorly supported alternative alleles (< 2 reads or < 5 %) were removed, as were SNPs where neither bulk had an index ≥ 0.3. Δ(SNP-index) was averaged in sliding windows of 2 Mb with a step of 10 kb (windows with fewer than 10 SNPs omitted). Confidence intervals were obtained by simulating [10,000] bulks of the given size in an [F2] population with the observed read depths (seed 12345); regions exceeding the [95] % interval were called QTL, adjacent significant windows less than 100 kb apart being merged.

8. Diversity Analysis

Describes genetic diversity of varieties from haplotype-block data: diversity indices per sample, PCA, hierarchical clustering, a neighbour-joining tree and the most variable blocks.

8.1 Input

CSV: column 1 = **Variety**, column 2 = **Sample ID**, columns 3+ = one column per haplotype block holding an integer allele/haplotype code. Code **0 is a real haplotype**; **−1 or blank = missing**.

8.2 Settings in the app

Setting	Default	Range / options	Effect
Number of clusters	5	2 – 20	Clusters from hierarchical clustering (Ward)
Top variable blocks	25	5 – 50 (step 5)	Number of most variable blocks in the block plots
Radar normalisation	on	on / off	Scales each index 0–1 in the radar plot (display only)

8.3 Internal (fixed) parameters

For each sample, p_i is the share of its non-missing blocks that carry code *i*.

Parameter	Value	Meaning
Richness	number of distinct codes	Richness_Normalized = richness / number of scored blocks
Load	sum of codes (missing counted as 0)	
Shannon	$-\Sigma\, p_i \ln(p_i + 10^{-10})$	Allelic_Entropy is the same in $\log_2$
Simpson (Gene diversity)	$1 - \Sigma\, p_i^2$	Gini–Simpson index
Effective alleles	$1 / \Sigma\, p_i^2$	
Heterozygosity	2f(1−f), f = sum of codes / (2 × scored blocks)	
PIC	$2\, p_{max}(1-p_{max})(1 - 2\, p_{max}(1-p_{max}))$	p_{max} = frequency of the commonest code
Nucleotide diversity	mean absolute difference between codes	Pairwise $\lvert code_a - code_b \rvert$
PCA	2 components	On the block-code matrix
Clustering	hierarchical, Ward linkage on pairwise distances	Dendrogram cut at the chosen number of clusters
Tree	neighbour-joining from Euclidean distances	

8.4 Outputs

Output	Content
`diversity_results.csv`	All indices per sample plus cluster and PCA scores
HTML summary and plots	Distribution of indices, PCA, dendrogram, NJ tree, radar and block plots

Methods text – Diversity Analysis

Diversity was assessed from [B] haplotype blocks in [N] samples with the Diversity Analysis module of TerraTools. For each sample, allelic richness, Shannon entropy, Gini–Simpson (gene) diversity, effective number of alleles, heterozygosity and polymorphism information content were computed from the frequency of each haplotype code among the sample's scored blocks (missing blocks ignored). Samples were ordered by principal component analysis, grouped into [5] clusters by Ward hierarchical clustering, and related by a neighbour-joining tree built from Euclidean distances.

9. GWAS

Genome-wide association with eight models (single-locus: OLS, MLM, EMMAX, GEMMA-style; multi-locus: FarmCPU, BLINK, mrMLM, FASTmrMLM). Only markers that pass a **genome-wide** threshold are listed as marker–trait associations (MTAs).

9.1 Input (three sources)

Source	Files	Notes
Rice 3K (built in)	none – choose accessions and a phenotype	3,024 accessions × 52,600 SNPs, reference IRGSP-1.0
VCF + phenotype	`genotypes.vcf / .vcf.gz` and phenotype CSV	Sample names must match; the VCF is converted to dosage 0/1/2 by the program
Ready file	one CSV: ID, Phenotype, then one column per SNP named Chr<N>_<position> (e.g. Chr1_1044946)	One row per accession; genotypes as dosage 0/1/2 (or centred −1/0/1); the encoding is detected, an ambiguous 0/1 file must be declared by the user

9.2 Settings in the app

Setting	Default	Range / options	Effect
Models	OLS + MLM	OLS, MLM, EMMAX, FarmCPU, GEMMA, BLINK, mrMLM, FASTmrMLM	Models run in one job; results are listed per model
Genotype encoding	auto	auto / dosage 0,1,2 / centred −1,0,1	How the genotype columns are read
Kinship	VanRaden	VanRaden / LD-pruned	Relationship matrix used by mixed models (Advanced)
LD r^2 (pruning)	0.2		Used for the pruned kinship and the PCs
LOCO	off	on / off	Leave-one-chromosome-out kinship
Number of PCs	0 per model (3 where a single value is used)	0 – 30	Population-structure covariates; **set per model** (OLS, MLM, EMMAX and GEMMA-style each have their own PC setting)
GEMMA test	Wald	Wald / LRT	Test statistic of the GEMMA-style model
FarmCPU iterations	10	1 – 100	Maximum iterations of FarmCPU
mrMLM search radius	20 kb	1 – 500 kb	Neighbouring markers combined in mrMLM / FASTmrMLM
mrMLM LOD cut-off	3		A QTN is reported only when LOD ≥ this
mrMLM principal components	3		Covariates for mrMLM
Significance rule	Bonferroni	Bonferroni / BH-FDR / simulated null	Genome-wide threshold (single-locus models)
Alpha	0.05		Family-wise or FDR level
Permutations / seed	1000 / 1		Used by the simulated-null threshold (residual permutation)
Plot options	Bonferroni line on; suggestive line and PVE optional		Display only

*QC thresholds (minor-allele frequency, missingness per marker and per sample, imputation) are **fixed** in this build (9.3); the QC tables* `gwas_qc_snps.csv`, `gwas_qc_samples.csv` *and* `gwas_qc_stages.csv` *record what each stage removed.*

9.3 Internal (fixed) parameters

Parameter	Value	Meaning
Marker QC	MAF ≥ 0 (none), marker missingness ≤ 1, sample missingness ≤ 1; duplicate positions kept	Defaults of this build: no MAF or missingness filtering (cut-offs 0 and 1). QC is run in stages (duplicate positions, call rates, monomorphic, MAF) and logged in the QC files; filter the file beforehand if you need cut-offs, and report them
Missing genotypes	mean imputation	Per marker, before kinship and testing
Kinship (VanRaden)	$K = ZZ' / (2 \Sigma p(1-p))$, $Z = G - 2p$	Missing contributes 0; PCs are computed from LD-pruned markers
MLM	REML eigen-decomposition; variance ratio d from the null model, s^2 re-estimated per marker	Written for this program (not a wrapper of TASSEL)
EMMAX	d and s^2 both fixed from the null model	Fastest mixed model; slightly conservative
GEMMA-style	d re-estimated per marker; Wald t or likelihood-ratio test	Re-implementation of the GEMMA test; the GEMMA program itself is not used
OLS	marker regression with the PCs set for OLS	No kinship; inflates in structured panels
FarmCPU	R package *rMVP* 1.4.6	Not re-implemented; settings above are passed through
BLINK	R package *GAPIT* 3.4.0	Not re-implemented
mrMLM / FASTmrMLM	R package *mrMLM* 5.0.1	Not re-implemented
Bonferroni	α / m (m = markers tested)	BH: Benjamini–Hochberg q ≤ α; simulated null: (1−α) quantile of the permuted minimum p (single-locus models)
Suggestive line	p = 1/m	Reference line in the plot only – **not** reported as MTA
Genomic inflation λ	median χ^2 / 0.456	Reported, not applied
PVE	model-specific: OLS marginal R^2; mixed models marginal; multi-locus conditional (joint)	Check the column header in the table
MTAs listed	genome-wide significant markers only (QTNs with LOD ≥ mrMLM LOD for mrMLM/FASTmrMLM)	No exploratory or suggestive MTAs

9.4 Outputs

Output	Content
`gwas_results.xlsx`	MTA table(s) per model, all-marker results, QC stages, settings
`gwas_mtas.csv`	Genome-wide significant MTAs (input for Genes Extraction)
`gwas_plots.zip`	Manhattan, QQ plots and PCA per model
QC files	`gwas_qc_stages.csv`, `gwas_qc_snps.csv`, `gwas_qc_samples.csv`

Methods text – GWAS

Genome-wide association analysis was carried out with the TerraTools GWAS module on [N] accessions and [M] SNPs after [QC description]. Missing genotypes were imputed with the marker mean. [Models]: a mixed linear model with a VanRaden kinship matrix ($K = ZZ'/2\Sigma p(1-p)$; [3] principal components as fixed covariates; REML variance ratio from the null model) was fitted for each marker; [FarmCPU (rMVP 1.4.6), BLINK (GAPIT 3.4.0) and mrMLM/FASTmrMLM (mrMLM 5.0.1) were run with default settings with [10] iterations / LOD ≥ [3] and a search radius of [20] kb]. The significance threshold was [Bonferroni, 0.05/m = [x]; $-\log_{10}(P) = [y]$]. Only SNPs above the threshold are reported as marker–trait associations. The genomic inflation factor was [λ = x]. Percentage of variance explained was calculated as [definition].

10. ET-GWAS

Extreme-trait GWAS (Gnanapragasam et al. 2024): lines in the high and low tails of a phenotype are compared with a random bulk drawn from the middle, using pooled allele counts and a binomial GLM. Input: a ready-made file with one row per line: ID, phenotype, then one column per SNP named Chr<N>_<position> with 0/1/2 dosages.

10.1 Settings in the app

Setting	Default	Range / options	Effect
Bulk size	20 % of lines	5 – 40 %	Fraction of lines in each extreme bulk
MAF filter	0.05		SNPs below this minor-allele frequency are not tested
Min. alleles per bulk	0 (= bulk size)		Minimum scored alleles per bulk at a SNP
Seed	1		Draws the random bulk from the middle of the distribution
Genomic control	on	on / off	Divides the statistics by λ (below)
Call MTAs by	FDR	FDR / suggestive p	How associations are declared
FDR level	0.05	0.01 – 0.2	BH q-value cut-off
Suggestive p	5×10^{-5}	10^{-3} … 10^{-6}	Used when the call mode is *suggestive*
Merge distance	100 kb	10 – 1000 kb	Significant SNPs closer than this form one region

10.2 Internal (fixed) parameters

Parameter	Value	Meaning
Minimum data	≥ 20 lines; each bulk ≥ 3 lines; the two bulks ≤ 80 % of the lines	Ties at the cut-off are kept
Random bulk	equal number of lines from the middle, chosen with the seed	
Test	binomial GLM (logit) on pooled ref/alt allele counts with the bulk rank low = 1, random = 2, high = 3	Pseudo-count 1; deviance statistic with 1 df
Genomic control	λ by regression through the origin of the sorted statistics (Devlin & Roeder 1999)	Statistics divided by λ when genomic control is on; λ is reported
Multiple testing	Benjamini–Hochberg q-values	
PVE	$2b^2f(1-f) / (2b^2f(1-f) + 2\,se^2\,N\,f(1-f))$	b = effect, f = allele frequency, N = lines

10.3 Outputs

Output	Content
`et_gwas_results.xlsx`	MTAs, all-SNP table, bulks, settings
`et_gwas_mtas.csv`, `et_gwas_all_snps.csv`, `et_gwas_bulks.csv`	Tables as CSV (bulks lists the lines in each bulk)
`et_gwas_plots.zip`	Bulk distribution, Manhattan, QQ, allele frequencies of top SNPs, PCA

Methods text – ET-GWAS

Extreme-trait GWAS (Gnanapragasam et al. 2024) was performed in TerraTools on [N] lines and [M] SNPs (MAF ≥ 0.05). The [20] % highest and lowest lines formed the high and low bulks and an equal number of lines sampled at random from the middle of the distribution (seed 1) formed the random bulk. Allele counts were pooled per bulk and tested with a binomial generalised linear model against the bulk rank; test statistics were corrected for inflation by genomic control (λ = [x]), and SNPs with Benjamini–Hochberg q < [0.05] were declared associated; significant SNPs less than [100] kb apart were merged into one region.

11. PyBSA-Seq (G-statistic BSA)

Bulked-segregant analysis from GATK4 variant tables using the G-statistic with a smoothed sliding window and simulation-based thresholds (Magwene et al. 2011). Run as a Python script behind the app; all settings below are passed to it.

11.1 Input

- Tab- or comma-separated GATK4 variant tables (`VariantsToTable`) for the **parents** and the **bulks**, with allele depths (AD) and genotype quality (GQ).
- Optional chromosome order file.

11.2 Settings in the app

Setting	Default	Range / options	Effect
Bulk sizes	430, 385	two integers	Number of individuals in each bulk (used for the simulation)
Population structure	F2	F2 / RIL / BC	Expected allele frequencies in the simulation
P-values	0.01, 0.01	two values	Significance level for the real data and for the simulation
Replications	10,000	1000 – 50,000	Number of simulation replicates
Sliding window	2,000,000 , 10,000	window, step (bp)	Window and step of the smoothing
Smoothing	51 , 3	window length, polynomial order	Savitzky–Golay filter of the G-statistic
Read length	100		Used by the read-depth model
Region of interest	−1 (none)	chromosome, start, end	Restricts the analysis to given regions
Misc: GQ cut-off, min. SNVs per window, extreme-read / mirror index	20, 5, 1		Variant and window filters
Parent is reference	no	yes / no	Whether one parent equals the reference genome
Adjust gap	off	on / off	Closes gaps between windows when smoothing

11.3 Internal (fixed) parameters

This script takes its thresholds from the settings above; no further constants are hard-coded in the app interface. For a methods section give the settings table and state the script/version of the TerraTools build. (Details of the G-statistic: Magwene et al. 2011; smoothing: Savitzky–Golay.)

11.4 Outputs

Output	Content
Report and plots	HTML report with G-statistic and smoothed profiles per chromosome, significance thresholds and QTL regions; tables in the output folder

Methods text – PyBSA-Seq

QTL regions were detected by bulked segregant analysis with the G-statistic (Magwene et al. 2011) as implemented in the PyBSA-Seq module of TerraTools. Bulks of [430] and [385] individuals from an [F2] population were compared using variants filtered for genotype quality ≥ [20] and at least [5] variants per window. The G-statistic was smoothed with a [2 Mb] sliding window (step [10 kb]) and a Savitzky–Golay filter (window length [51], order [3]); significance thresholds (P = [0.01]) were obtained from [10,000] simulation replicates.

12. Genes Extraction

Lists the genes (and optionally sequences) that lie in QTL intervals, around GWAS SNPs, or at named markers, using a GFF3 annotation.

12.1 Input

File	Content
Regions (one of)	**GWAS table**: `SNP, Chr, Position` (optional `P_value`); **QTL intervals**: `Chromosome, Start, End` (also QTL_ID / QTL_Start / QTL_End, Left/Right marker names, or a qtl_significant table where marker names carry the position); **marker names only**. Excel files are read sheet by sheet
Annotation	GFF3 with *gene* lines (all attributes are kept as columns)
Genome FASTA (optional)	Needed only for sequences; genes on the minus strand are reverse-complemented

Chromosome names are matched across notations: Chr01, chr1, S1 and 1 are the same chromosome.

12.2 Settings in the app

Setting	Default	Range / options	Effect
Window (kb)	100	≥ 0	Applied on **each side of a point** (GWAS SNPs); 0 = only the gene that contains the SNP. Intervals are used as given
P-value limit	none	optional	Keeps only SNPs with P ≤ limit
Match mode	inside	inside / overlap	*inside*: the whole gene lies in the region; *overlap*: any overlap counts

12.3 Internal (fixed) parameters

Parameter	Value	Meaning
Maximum regions	20,000	Larger inputs are refused
Attribute columns	up to 40	Further GFF3 attributes are dropped
Excel cell limit	32,000 characters	Longer values (e.g. sequences) are truncated in the Excel file; the FASTA file is complete

12.4 Outputs

Output	Content
Gene table (Excel)	Region, gene ID, coordinates, strand and all annotation attributes; sheet *Gene_ID_list*
`Gene_ID_list.txt`	One gene ID per line – input of Gene Prioritization
`Gene_sequences.fasta`	If a FASTA file was supplied

Methods text – Genes Extraction

Genes located in the [QTL support intervals / ±[100] kb windows around significant SNPs] were extracted from the [IRGSP-1.0] GFF3 annotation with the TerraTools Genes Extraction module; genes lying [entirely inside / partly overlapping] the regions were retained ([n] regions, [g] genes).

13. Gene Prioritization

Ranks candidate genes by combining genetic evidence from your mapping result with public functional evidence (Ensembl Plants, Gramene, Plant Reactome, NCBI / PubMed). **Internet access is required.**

13.1 Input

- Species: rice (*Oryza sativa*, IRGSP-1.0, RAP-DB IDs; MSU IDs cannot be mapped), wheat (IWGSC RefSeq v1.0) or maize (B73 NAM-5.0). Chickpea is disabled.
- Trait text (e.g. “grain size”) used for the literature search.
- GWAS or QTL file with column mapping (the app asks which column is gene, chromosome, position, start, end, p, adjusted p, LOD, QTL ID, marker) **or** a gene list.

13.2 Settings in the app

Setting	Default	Range / options	Effect
Species	rice	rice / wheat / maize	Annotation and ortholog sources
Trait	–	free text	Keywords for literature and function matching
Window (kb)	100		Genes within this distance of a SNP are candidates
Refresh cache	off	on / off	Re-downloads evidence instead of using the cache (30 days)

13.3 Internal (fixed) parameters

Parameter	Value	Meaning
Evidence weights	association 0.30, function 0.25, pathway 0.20, ortholog 0.15, literature 0.10	Fixed weights of the five categories
Score	$S = \Sigma\, w_c\, E_c \,/\, \Sigma\, w_c \times 100$, over the categories with evidence	Missing evidence is **not** counted as negative; the score is not a probability of causality
Genetic evidence	locus = 0.6 significance + 0.4 proximity; gene = 0.75 best locus + 0.25 support from independent loci	Full marks at $-\log_{10} p = 8$ or LOD = 10; two loci give 0.75, three 0.875
Evidence codes	experimental 1.0, curated 0.8, other 0.6	Strength of annotation evidence
Match kind	direct 0.6, related 0.4	Trait term in the function text
Ortholog types	1:1 = 1.0, 1:many = 0.8, many:many = 0.7	
Limits	600 candidate genes; literature for the top 80; 15 publications per gene; 8 workers	Larger lists are truncated by genetic evidence
Network	30 s timeout, 4 retries, back-off 1.5 s, circuit breaker after 8 failures	Failed look-ups are cached for 7 days

13.4 Outputs

Output	Content
Ranking table	Gene, locus, genetic evidence, five category scores, final score and rank, evidence text, publications (Excel/CSV and on-screen)

Methods text – Gene Prioritization

Candidate genes within [100] kb of [QTL / GWAS signals] were ranked with the TerraTools Gene Prioritization module. Each gene received scores for genetic association (significance and proximity of the signal), functional annotation, pathway membership (Plant Reactome), orthology to characterised genes (Gramene) and literature support (PubMed) for the trait “[trait]”, which were combined into a weighted score (weights 0.30, 0.25, 0.20, 0.15 and 0.10) normalised over the evidence available for the gene. Annotation data were retrieved on [date]. The score indicates the strength of supporting evidence and not the probability that a gene is causal.

14. Clustering and Genomic Selection split

Groups genotypes by SNP data (PCA + k-means) and splits the population into a **training** and a **testing** set that keeps every cluster represented – a representative split for genomic prediction.

14.1 Input

CSV or TSV: first column = sample IDs, other columns = SNP markers (0/1/2); **−1 = missing**.

14.2 Settings in the app

Setting	Default	Range / options	Effect
Number of clusters (K)	5	2 – 20	Clusters in k-means (reduced automatically to at most n − 1)
Test size	0.2	0.1 – 0.8	Fraction of each cluster placed in the testing set

14.3 Internal (fixed) parameters

Parameter	Value	Meaning
Missing values	−1 → missing → column mode (0 if a column has no data)	Simple mode imputation
PCA	all components	*random_state* 42
k-means	on the PCA scores, *n_init* 20, *random_state* 60	
Split	stratified shuffle split on the cluster label, *random_state* 42	If a cluster has fewer than 2 members or n < 4, a plain random split is used instead
Plot	PC1 vs PC2, 300 dpi	Colours = clusters

14.4 Outputs

Output	Content
`Training_data_Cluster.csv`, `Testing_data_Cluster.csv`	The two sets (with cluster labels)
`cluster_membership.csv`, `cluster_plot.png`	Cluster of every sample; PCA plot

Methods text – Clustering and training/testing split

Missing genotypes were replaced by the most frequent genotype of the marker. Principal component analysis was applied to the genotype matrix and samples were grouped into [K] clusters by k-means (20 random starts, seed 60) on the principal component scores. Each cluster was split into training ([80] %) and testing ([20] %) samples by stratified random sampling (seed 42) so that all clusters are represented in both sets.

15. Additive Dominance (relationship matrices)

Builds the additive (A) and dominance (D) genomic relationship matrices from a SNP matrix, for use in genomic prediction and mixed models.

15.1 Input

CSV or TSV (separator detected): samples in rows, first column = sample ID, other columns = markers coded 0/1/2. At least 2 samples.

15.2 Settings in the app

Setting	Default	Range / options	Effect
Missing-value imputation	mean	mean / median / zero	Applied per marker before the matrices are built
Matrices	additive and dominance	either or both	Which matrix files are written

15.3 Internal (fixed) parameters

Parameter	Value	Meaning
Monomorphic markers	removed (variance 0)	The number with MAF > 0.05 is reported for information only; no MAF filter is applied
Coding	0/1/2 recoded to −1/0/1	Needed by the R functions
Additive matrix	*rrBLUP::A.mat* (rrBLUP 4.6.3)	VanRaden-type marker-based relationship matrix (Endelman 2011)
Dominance matrix	*sommer::D.mat* (sommer 4.4.6)	Covarrubias-Pazaran 2016
Previews	heat map max. 60 × 60; table preview 100 rows; histogram 60 bins	Display only; the CSV files are complete

15.4 Outputs

Output	Content
`Additive_Matrix.csv`, dominance matrix CSV	Square matrices (samples × samples) with sample names
Plots	Heat maps and histogram of the matrix values

Methods text – Additive Dominance

Monomorphic markers were removed and missing genotypes were replaced by the marker [mean]. The additive relationship matrix was computed with rrBLUP::A.mat (rrBLUP 4.6.3; Endelman 2011) and the dominance relationship matrix with sommer::D.mat (sommer 4.4.6; Covarrubias-Pazaran 2016) from [N] samples and [M] markers.

16. BreedSight Genomic Prediction

Genomic prediction with a deep neural network (DNN) and random forest combined as an ensemble, benchmarked against Lasso, RBLUP and GBLUP by cross-validation. Runs in Python (TensorFlow, scikit-learn).

16.1 Input

Four CSV files: training phenotypes, training genotypes, test phenotypes, test genotypes. Genotypes coded 0/1/2 with **no missing values**; sample names must match between each phenotype and genotype pair.

16.2 Settings in the app

Setting	Default	Range / options	Effect
Outer folds	5	2 – 10	Cross-validation folds on the training data
Heritability	0.82	0 – 1	Used only when interpreting the accuracy; does not change the model
Epochs	300		Maximum DNN epochs (early stopping applies)
Batch size	32		
Learning rate	0.001		Adam learning rate
Ensemble weight α	0.5	0 – 1	Prediction = α · DNN + (1 − α) · RF
L2 penalty	0.01		Weight regularisation of the DNN
Dropout	0.3		DNN dropout rate
RF trees / max depth	200 / 15	depth 0 = unlimited	Random-forest size
Use PCA	on	on / off	Reduces the feature matrix to the given components
PCA components	50		
Feature selection (RFE)	on	on / off	Recursive feature elimination to the chosen number of features
RFE features	100		
Use raw genotypes	off	on / off	Use the markers instead of the kernel features below

16.3 Internal (fixed) parameters

Parameter	Value	Meaning
Random state	90	Seeds the folds and the models
Cross-validation	K-fold with shuffling	Metrics are computed on out-of-fold predictions
Features	G = centred X X′ / p, concatenated with the normalised G□G	G□G adds an epistasis-like (squared) relationship term. Raw genotypes only if selected
Feature selection	RFE with a 50-tree random forest	Down to the chosen number of features
DNN	256 – 128 – 64 units, batch normalisation, dropout, LeakyReLU (0.1), He-normal initialisation, L2, linear output	Huber loss (delta 1); Adam with clip value 0.5
Training control	early stopping (patience 15); learning-rate reduction ×0.5 (patience 10, minimum 1e−7)	
Benchmarks	Lasso (α 0.1, 10,000 iterations); RBLUP (ridge, α 1.0); GBLUP (kernel ridge, RBF, α 1.0)	Fitted on the same folds
Final model	retrained on all training data, 300 epochs	Predicts the test set

16.4 Outputs

Output	Content
Metrics	Per fold and mean prediction accuracy (correlation and error measures) for the ensemble and the benchmarks
Predictions	Test-set predictions (CSV) and observed-vs-predicted plots

Methods text – BreedSight Genomic Prediction

Genomic prediction was performed with the BreedSight module of TerraTools (Python, TensorFlow, scikit-learn) using [n] training and [m] test genotypes with [p] markers. The feature matrix combined the centred genomic relationship matrix with its element-wise square [after PCA to [50] components and recursive feature elimination to [100] features]. A DNN (256-128-64 units; batch normalisation, dropout [0.3], L2 [0.01]; Huber loss; Adam, learning rate [0.001]; early stopping) and a random forest ([200] trees, depth [15]) were combined as α·DNN + (1−α)·RF with α = [0.5]. Accuracy was the Pearson correlation between predicted and observed values in [5]-fold cross-validation (seed 90); Lasso, ridge regression (RBLUP) and kernel ridge regression (GBLUP) served as benchmarks.

17. Optimal Cross Selection

Chooses which crosses to make: it maximises the expected gain of the offspring while limiting inbreeding (coancestry), and shows the whole trade-off curve (frontier).

17.1 Input

- **Criterion CSV**: columns `Genotype` and `Criterion` (for example GEBV).
- **Kinship CSV**: square relationship matrix; genotype names in the first column and in the header. The diagonal is set to 0.5.

17.2 Settings in the app

Setting	Default	Range / options	Effect
Number of crosses	20	1 – 500	Crosses in the plan
Max. crosses per parent	4	1 – 8	Limits how often one parent is used
Penalty degree (θ)	30°	0 – 90°	Weight between gain and inbreeding: 0° = gain only (truncation-like), 90° = inbreeding only

17.3 Internal (fixed) parameters

Parameter	Value	Meaning
Contribution	$x_i = n_i / (2 \times \text{crosses})$	n_i = number of times parent *i* is used
Gain	mean criterion of the parents (weighted by contribution)	
Group coancestry	$c = x'Cx$	C = kinship matrix
Progeny inbreeding	f = mean kinship of the mated pairs	Coancestry term used = c + f
Objective	$J = \cos\theta \cdot \text{gain}_{\text{scaled}} - \sin\theta \cdot \text{coancestry}_{\text{scaled}}$	Both scaled 0 – 1 between random mating and truncation selection
Optimiser	simulated annealing, temperature 0.05 → 0.0005	Frontier: 40,000 iterations at each angle (0°, 1°, 5°, 10° … 90°); chosen angle: 150,000 iterations from several starts
Starting plan	best parents paired at random, no selfing, no repeated pairs	
Seed	1	Reproducible plans

17.4 Outputs

Output	Content
`ocs_crosses.csv`	The cross list (parent 1, parent 2)
`ocs_contributions.csv`	Contribution of each parent
`ocs_frontier.csv`, `ocs_frontier.png`	Gain vs coancestry at every angle, with the truncation-selection comparison

Methods text – Optimal Cross Selection

Crosses were selected by optimal cross selection in TerraTools: [20] crosses, each parent used in at most [4] crosses, maximising J = cosθ · gain − sinθ · coancestry with θ = [30]°, where gain is the mean [GEBV] of the parents and coancestry is the group coancestry x′Cx plus the mean kinship of the mated pairs (both scaled between random mating and truncation selection). The plan was optimised by simulated annealing (150,000 iterations, several starts, seed 1). The achieved gain and coancestry are [x] and [y], compared with [x′] and [y′] under truncation selection of the same number of parents.

18. Heterosis

Predicts the performance of hybrids from parental genotypes with a deep neural network trained on a phenotyped training set, and ranks the F1 hybrids (mid-parent and best-parent heterosis). Runs in Python (TensorFlow).

18.1 Input

- **Training phenotype CSV**: column 1 = ID, column 2 = the trait.
- **Training genotype CSV** and **parent genotype CSV**: the same markers, 0/1/2.
- **Test genotype CSV** (optional): other lines to predict.

18.2 Settings in the app

Setting	Default	Range / options	Effect
Top F1 hybrids to list	5	1 – 50	
Top parents to list	5	1 – 30	
Folds (k)	5	2 – 10	Cross-validation folds
Epochs	100	10 – 500	
Batch size	16	8 – 128	
Dropout	0.2	0 – 0.5	
L1 / L2 penalty	0.001 / 0.001		Weight regularisation

18.3 Internal (fixed) parameters

Parameter	Value	Meaning
Seed	42	Python, NumPy and TensorFlow
Scaling	StandardScaler fitted on the training genotypes	Applied to training, parents and test genotypes
Network	512 – 512 – 256 – 128 – 64 – 32 – 16 – 8 units with L1/L2 and dropout; one ReLU output	Loss = MSE; Adam with default learning rate
Cross-validation	K-fold, shuffled, *random_state* 60	MSE, RMSE, R^2 and Pearson r averaged over folds
Final model	trained on all training data	
Hybrid genotype	mean of the two standardised parent genotype vectors	All pairwise parent combinations are scored
MPH	$(GEBV_{hybrid} - \text{mean(training phenotype)}) / \text{mean(training phenotype)} \times 100$	The reference is the training-set **mean**, not the parental values
BPH	$(GEBV_{hybrid} - \max(\text{training phenotype})) / \max(\text{training phenotype}) \times 100$	The reference is the training-set **maximum**
SCA list	mean of the two parental GEBVs	As produced by the program

18.4 Outputs

Output	Content
Top F1 hybrids, top parents, all hybrids	Predicted GEBV, MPH and BPH for every combination (CSV)
Test predictions	Predicted values for the optional test lines
Metrics	Cross-validation MSE, RMSE, R^2 and Pearson r (mean over folds)

Interpreting MPH and BPH

MPH and BPH here are measured against the *training population* (its mean and its best value), not against the actual parents of each hybrid. State this definition whenever you report them.

Methods text – Heterosis

Hybrid performance was predicted with a deep neural network (512-512-256-128-64-32-16-8 units, L1/L2 = 0.001, dropout [0.2], Adam, MSE loss; TensorFlow) trained on [n] phenotyped lines with [p] standardised markers ([100] epochs, batch [16], seed 42). Predictive ability in [5]-fold cross-validation was r = [x] (R^2 = [y]). Genotypes of [h] F1 hybrids were generated as the mean of the standardised parental genotypes and the model was applied to them. Mid-parent and best-parent heterosis were expressed relative to the training-set mean and maximum, respectively.

19. Reporting a result

19.1 Checklist – what to write down for each tool

Details and definitions are in the chapter of each tool; the minimum is:

Tool	Settings and data	Fixed values to state	Results to report
Phenotypic (2)	Design, genotypes, checks, environments, reps/blocks, traits, k	Federer/lmer model; H^2 formulas; k-means nstart 25 / Ward.D2	Adjusted means, variance components, H^2 (plot and entry-mean), PCA %, clusters
Mendelian (3)	Population, p threshold, missing codes	Chi-square 1:1 (RIL) / 1:2:1 (F2); ≥ 5 genotypes	Markers tested, distorted (n, %), by chromosome
Convert Alleles (4)	Separator, columns	Coding rules, excluded categories	Markers read / coded / excluded by reason, % missing
Linkage (5)	Population, missing, max r, order, method, chi-square	Kosambi, error 1 %, group LOD 5, weak-link LOD 3, poor fit 10 cM, given order 3 LOD	Markers, positions, length (cM), spacing, gaps, breaks, removed by reason
QTL (6)	Method, threshold mode/value, permutations, α, peak p, drops, cofactors, window, hold-out	Seed 42; flanking-marker fill; cofactor p 0.01/0.05; MQM and ML constants	Per QTL: chr, peak, interval (cM, bp), LOD, genome-wide P, PVE, effect ± SE
QTL-Seq (7)	Mode, population, bulk sizes, depth/quality filters, window, replicates, level	Seed 12345; null model; per-window intervals (no multiple-testing correction)	Regions (bp), peak Δ(SNP-index), SNPs per region
Diversity (8)	Clusters, top blocks	Index definitions; Ward; NJ	Indices per group, PCA, clusters
GWAS (9)	Source, models, encoding, kinship, PCs, rule, α	QC defaults, imputation, REML details, R packages and versions	Threshold value, λ, MTAs (SNP, chr, pos, P, effect, PVE), model
ET-GWAS (10)	Bulk %, MAF, seed, FDR or suggestive, merge distance	GLM, genomic control, BH, PVE formula	Bulks (n), λ, regions and q-values
PyBSA-Seq (11)	All settings of the table	Window, smoothing, simulation	Regions, thresholds
Genes Extraction (12)	Window, match mode, p limit, annotation file (name, version)	Region and attribute limits	Regions, genes per region, gene list
Gene Prioritization (13)	Species, trait text, window, retrieval date	Weights; evidence rules; limits	Top-ranked genes with evidence, not probabilities
Clustering (14)	K, test size	Imputation; seeds 42 / 60	Cluster sizes; train/test numbers
Additive Dominance (15)	Imputation, matrices	rrBLUP / sommer functions	Dimensions; markers removed
BreedSight (16)	All hyper-parameters, folds	Random state 90; architecture; benchmarks	Accuracy per fold and mean, benchmarks
OCS (17)	Crosses, max per parent, θ	Objective; annealing; seed 1	Gain, coancestry, truncation comparison, cross list
Heterosis (18)	Hyper-parameters, folds	Seed 42; architecture; MPH/BPH definition	CV accuracy; top hybrids

Always: TerraTools installer version, R/Python component versions (section 1.4), the input file names/dates, the results folder, and numbers of individuals and markers before and after each filtering step.

19.2 Convert Alleles and Linkage Mapping

Replace the values in [brackets] with your own; delete the sentences that do not apply (for example the chi-square sentence if the filter was off).

Genotype coding and linkage map construction

Raw genotype calls of [N] [F2 / RIL] individuals at [M] markers were coded against the parental genotypes with the Convert Alleles module of TerraTools (0 = homozygous for parent 1, 1 = heterozygous, 2 = homozygous for parent 2, −1 = missing or inconsistent with the parents). Markers with a missing, heterozygous or identical parental call were excluded ([x], [y] and [z] markers, respectively). Linkage maps were constructed with the TerraTools Linkage Map Creator in R 4.6.1. Markers with more than [30] % missing data and monomorphic markers were removed[, as were markers showing segregation distortion (chi-square test, α = [0.05][, Bonferroni-corrected])]. Markers with identical genotype patterns were represented by one locus. Linkage groups corresponded to the [12] chromosomes of the marker file; a marker was removed if it was more strongly linked (independence LOD ≥ 5) to another chromosome, if its strongest LOD with any marker of its own chromosome was below 3, or if it lengthened the map by more than 10 cM. Within each chromosome, marker order was searched by minimising the sum of adjacent recombination frequencies and refined by likelihood-based permutation of sliding windows of markers (following the logic of JoinMap and Lep-MAP3); the physical order given by the marker names was kept unless the data-driven order increased the log-likelihood by more than 3 LOD. Recombination frequencies were estimated by multipoint maximum likelihood with a hidden Markov model (EM algorithm; genotyping error rate 1 %) and converted to centiMorgans with the Kosambi mapping function; neighbouring markers with a recombination fraction ≥ [0.45] were considered unlinked and the interval was set to 50 cM. The final map comprised [n] markers at [u] unique positions on [k] linkage groups, with a total length of [L] cM and a mean marker spacing of [s] cM.

19.3 QTL Mapping

QTL analysis

QTL analysis was performed with the TerraTools QTL Mapping module in R 4.6.1 using [n] individuals, [m] markers and the trait [trait]. Missing genotypes ([x] %) were replaced by the most probable genotype given the nearest flanking markers (Kosambi map function). [SIM] Single-marker regression was used to calculate LOD = $(n/2)\,\log_{10}(RSS_0/RSS_1)$ at every marker (Haley & Knott 1992). [CIM] Composite interval mapping (Zeng 1994) added up to [5] background markers selected by forward selection and backward elimination (p to enter 0.01, p to remove 0.05); cofactors within [10] cM of the tested position were excluded from the model. The genome-wide LOD threshold was obtained from [1000] permutations of the phenotype (α = [0.05]; Churchill & Doerge 1994) and was [3.54] [for CIM, the permutation null was taken from single-marker regression]. A QTL was declared at a local LOD maximum above the threshold with a nominal P ≤ [0.05]; peaks on the same chromosome were treated as separate QTL when the LOD curve fell by at least [2] between them. The support interval was the [1.5]-LOD drop interval expanded to the nearest flanking markers. The additive effect, its standard error and the percentage of phenotypic variance explained (PVE) were estimated in the regression model [including the cofactors]. Confidence intervals in cM were converted to base pairs by linear interpolation between markers with known physical positions. The random seed was 42.

If you also used the machine-learning methods:

RF / SVR / XGBoost (optional paragraph)

As complementary, non-parametric approaches, random forest (500 trees; Breiman 2001), support-vector regression (radial kernel) and gradient boosting (XGBoost; 100 rounds, maximum depth 3, learning rate 0.1, subsample 0.8, column sample 0.8; Chen & Guestrin 2016) were fitted on all markers; marker importance (permutation importance, or gain for XGBoost) was rescaled to a LOD-like score and used only to confirm regions found by regression-based mapping. Regions detected by at least [2] of [3] methods were reported as common QTL.

19.4 Suggested results table for QTL

QTL ID	Chr	Peak marker	Peak (cM)	Interval (cM)	Interval (Mb)	LOD	Genome-wide P	PVE (%)	Additive ± SE
QTL_1.1	1	[marker]	[ ]	[start – end]	[start – end]	[ ]	[ ]	[ ]	[±]
QTL_5.1	5	[marker]	[ ]	[start – end]	[start – end]	[ ]	[ ]	[ ]	[±]

Add a footer: "Threshold: genome-wide permutation LOD = [x] (1000 permutations, α = 0.05). PVE from the regression model; additive effect in [trait unit] (positive: the allele from [parent 2] increases the trait)."

19.5 References to cite

Cite the method that produced your result. Check the details against the originals before submission.

- **Linkage / QTL.** Kosambi DD (1944) The estimation of map distances from recombination values. *Annals of Eugenics* 12: 172–175. Van Ooijen JW (2011) Multipoint maximum likelihood mapping in a full-sib family of an outbreeding species. *Genetics Research* 93: 343–349. Rastas P (2017) Lep-MAP3. *Bioinformatics* 33: 3726–3732. Lander ES, Botstein D (1989) *Genetics* 121: 185–199. Haley CS, Knott SA (1992) *Heredity* 69: 315–324. Churchill GA, Doerge RW (1994) *Genetics* 138: 963–971. Zeng Z-B (1994) *Genetics* 136: 1457–1468. Jansen RC, Stam P (1994) *Genetics* 136: 1447–1455. Broman KW, Sen Ś (2009) *A Guide to QTL Mapping with R/qtl*. Springer.
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- **Bulked-segregant analysis.** Takagi H et al. (2013) QTL-seq: rapid mapping of quantitative trait loci in rice by whole genome resequencing of DNA from two bulked populations. *Plant J.* 74: 174–183. Magwene PM, Willis JH, Kelly JK (2011) The statistics of bulk segregant analysis using next generation sequencing. *PLoS Comput. Biol.* 7: e1002255. Savitzky A, Golay MJE (1964) *Anal. Chem.* 36: 1627–1639.
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- **Genomic selection.** Endelman JB (2011) Ridge regression and other kernels for genomic selection with R package rrBLUP. *Plant Genome* 4: 250–255. Covarrubias-Pazaran G (2016) Genome-assisted prediction of quantitative traits using the R package sommer. *PLoS ONE* 11: e0156744. Meuwissen THE (1997) Maximizing the response of selection with a predefined rate of inbreeding. *J. Anim. Sci.* 75: 934–940. Kinghorn BP (2011) An algorithm for efficient constrained mate selection. *Genet. Sel. Evol.* 43: 4.
- **Diversity.** Saitou N, Nei M (1987) The neighbor-joining method. *Mol. Biol. Evol.* 4: 406–425.

20. Worked example (training kit)

The training kit (`training_demo/`) uses 176 rice RILs of the BSP2 population with 334 informative markers on 12 chromosomes. The genotypes are real; the allele letters, the 11 faulty markers and the three traits are simulated (seed 2026), so the answer is known.

Step	Settings	Result
Convert Alleles	separator ":" (auto); Marker / Chromosome / P1 / P2 columns auto-detected	345 markers read, 334 coded, 11 excluded (3 invalid parent, 4 heterozygous parent, 4 non-segregating)
Linkage Mapping	RIL; missing 0.30; max r 0.45; best marker order; chi-square off	334 markers on 12 chromosomes, total ≈ 1,363 cM
QTL – SIM, Yield	fixed LOD 3	4 QTL: chr1 LOD 41.9 (PVE 66.6 %), chr5 LOD 12.3, and two ghost peaks (chr4 LOD 3.2, chr7 LOD 3.5)
QTL – CIM, Yield	permutation, 1000, α 0.05 → threshold LOD 3.54; 5 cofactors, 10 cM	2 QTL: chr1 LOD 62.4 (effect 2.07), chr5 LOD 32.4 (effect 1.14) – planted: +2.0 and +1.2
QTL – CIM, Plant_Height	permutation, threshold 3.56	chr9 LOD 43.8 (effect 4.30), chr5 LOD 18.8 (effect 2.33) – planted: +4.0 and +2.5
QTL – CIM, No_QTL_Trait	permutation, threshold 3.61	0 QTL (negative control)

The lesson of the example: with a fixed LOD of 3 the scan reported ghost peaks; the permutation threshold removed them and left exactly the planted QTL, with effects close to the true values.

21. Troubleshooting and good practice

Tool	Symptom	Likely cause and remedy
All	“File not found” / empty results	Path with special characters, file open in Excel, or wrong sheet; keep input and output folders on a local disk
All	Few samples match	IDs differ between files (spaces, case, prefixes). Make the IDs identical
Convert Alleles	“No offspring genotype columns could be read”	Wrong allele separator (A:G needs “:”, AG needs “none”), or Marker/P1/P2 columns mis-assigned
Convert Alleles	Many Non-segregating markers	Parent columns swapped with an individual, or closely related parents
Linkage	Map far longer than expected	Genotyping errors: read Marker Fit, lower the missing threshold, check 50 cM linkage breaks
Linkage	Many “Linked to another chromosome”	Chromosome labels of the file may be wrong
QTL	“Only n matching samples”	Genotype and phenotype IDs differ
QTL	No QTL found / extra peaks	Compare the full-scan maximum LOD with the threshold; use permutation threshold and CIM
QTL	Intervals file empty	Marker names carry no physical position and no linkage map was given
GWAS	Strong inflation (λ > 1.2) in OLS	Population structure: use MLM/EMMAX or add PCs
GWAS	No MTAs	Genome-wide threshold not reached – report the top signals as such, not as MTAs
ET-GWAS	Refused to run	Fewer than 20 lines, or bulks larger than 80 % of the lines
QTL-Seq / PyBSA	Empty plot	Too few SNPs pass the depth filter; lower min. depth or window size
Genes Extraction	No genes	Chromosome names differ from the GFF3 (check notation), or the window is 0
Gene Prioritization	Slow or partial result	Needs internet; the network limit/circuit breaker may have stopped; retry later (cached evidence is reused)
BreedSight / Heterosis	Memory error or very slow	Reduce epochs, markers (PCA/RFE) or hybrids; check TensorFlow is installed
OCS	Kinship file rejected	Not square, or names in header and first column differ

- Keep the **input files**, the settings and the result folder together (chapter 19).
- Run a negative control or a permutation test before trusting a peak.
- Report the threshold **value**, not only the method used to get it.